

AWS Fargate

1. What is AWS Fargate?

AWS Fargate is a serverless compute engine for containers. It allows you to run containerized applications without managing EC2 servers or clusters.

It integrates with:

- Amazon ECS (Elastic Container Service)
- Amazon EKS (Elastic Kubernetes Service)

With AWS Fargate, you only define:

- CPU & Memory
- Container Image
- Networking configuration
- IAM permissions

Analogy: Fargate is like AWS Lambda's big sibling — designed for long-running, container-based workloads that need more control, networking, and scaling.

2. Why Use Fargate?

Feature	Value	Limitation
No servers to manage	Fully serverless	No GPU support
Cost-efficient	Pay only for consumed CPU & RAM	Can be expensive at massive scale
Secure by design	Task-level isolation (kernel sandboxing)	Limited host customization

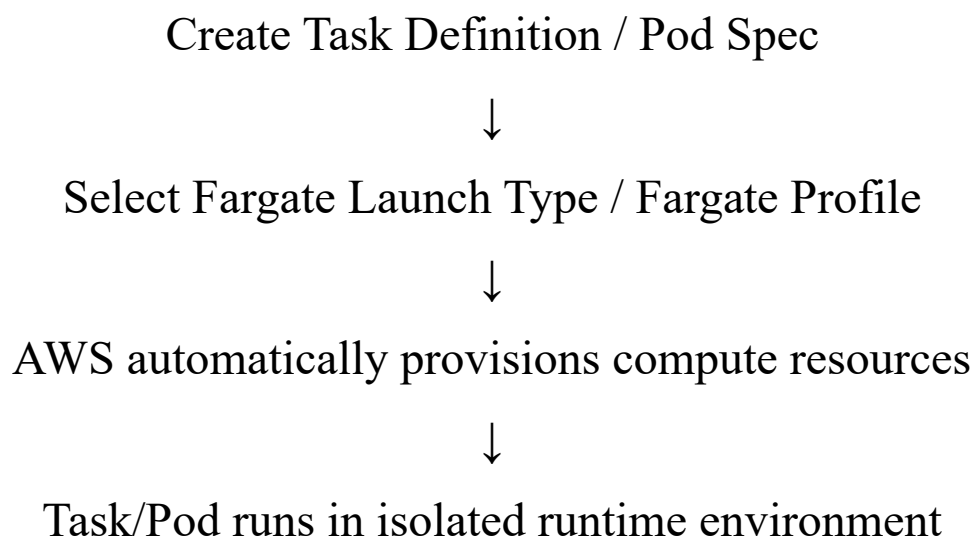
Feature	Value	Limitation
Auto scaling	Scale tasks, not EC2 instances	ENI/IP quota can bottleneck
AWS ecosystem integration	Works with IAM, CloudWatch, ECR, ALB, EFS	No daemon sets (EKS)

3. Core Concepts

Concept	ECS Term	EKS Term	Meaning
Blueprint	Task Definition	Pod Spec	Defines how a container should run
Running Instance	Task	Pod	The active execution
Long-running Group	Service	Deployment	Ensures desired number of tasks
Placement Control	Launch Type	Fargate Profile	Determines whether workload runs on Fargate
Scaling Unit	Service Auto Scaling	HPA (Horizontal Pod Autoscaler)	Defines how workloads scale



4. How Fargate Works (Flow)



How Fargate Works (Flow)



5. Common Use Cases

- Microservices and backend APIs
- Batch workloads and scheduled jobs
- CI/CD runners (Jenkins agents, GitHub runners)
- Event-driven workloads (SNS, SQS, EventBridge)
- Web applications behind ALB

6. ECS Launch Types: Fargate vs EC2

Feature	ECS on EC2	ECS on Fargate
Infrastructure	User manages EC2	Fully managed
Scaling	Scale servers	Scale tasks
Pricing	Pay for instances	Pay per CPU + Memory
Isolation	Shared OS	Task-level isolation
Startup Time	Minutes (boot instance)	Seconds (task starts quickly)
Best for	GPU, spot, custom workloads	Serverless workloads

7. Using Fargate with ECS

Requirements:

- VPC Networking
- IAM Task Role
- Logging (CloudWatch / FireLens)

Supported features:

- ALB / NLB
- Auto Scaling
- EFS Volumes

8. Using Fargate with EKS

- Requires a **Fargate Profile** (namespace + labels).
- Best for Kubernetes teams that don't want to manage worker nodes.

9. Pricing Model

Charged by:

- vCPU-seconds
- Memory (GB-seconds)
- Optional: EFS storage

 Minimum billing time: 1 minute

10. AWS Fargate Limits

Resource	Limit
vCPU per task	0.25 – 64
Memory per task	0.5 – 120 GB
Ephemeral storage	20 GB (up to 200 GB since 2023)
Networking mode	awsvpc only
ENI per task	1
Max tasks per service	500 (soft limit, can be raised)

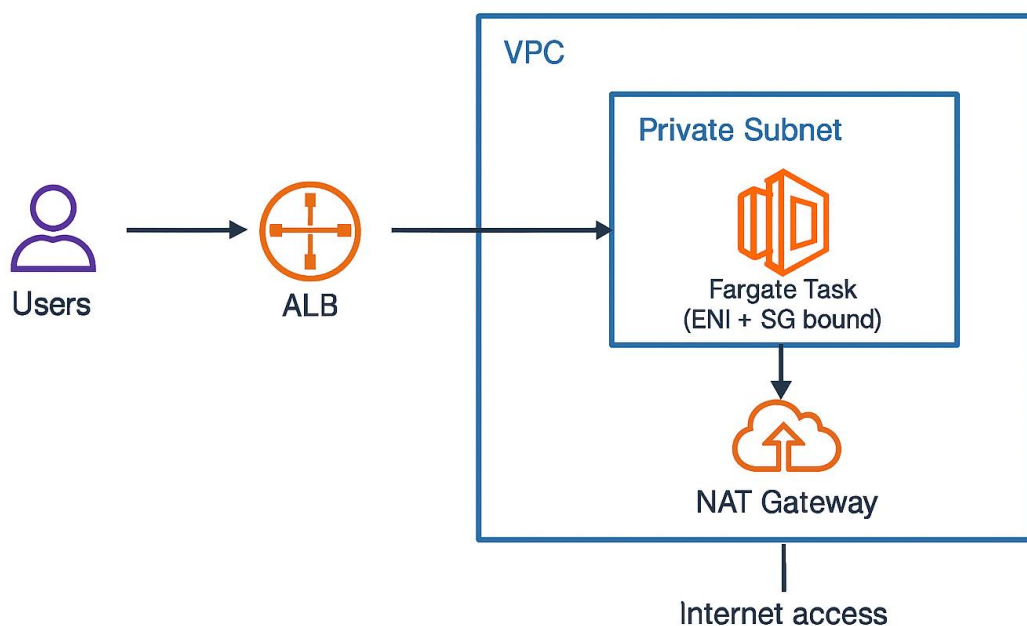
11. Networking Model

- Each task gets its own **Elastic Network Interface (ENI)**.
- Security Groups can be applied per task.

Supports:

- Private subnets + NAT
- VPC Flow Logs
- Service-to-service networking

Pitfall: Each ENI consumes IP addresses from your subnet. Plan CIDR ranges carefully. Scaling can fail if ENI quota is exceeded.



12. Storage Support

Storage Type	Supported	Notes
Ephemeral Storage	✓ Yes	Temporary runtime
EFS	✓ Yes	Persistent storage
EBS / FSx	✗ No	Not supported
S3	✓ API access only	Not mountable as volume

13. Security Features

- IAM Roles assigned at task level
- Secrets from:
 - AWS Systems Manager Parameter Store
 - AWS Secrets Manager

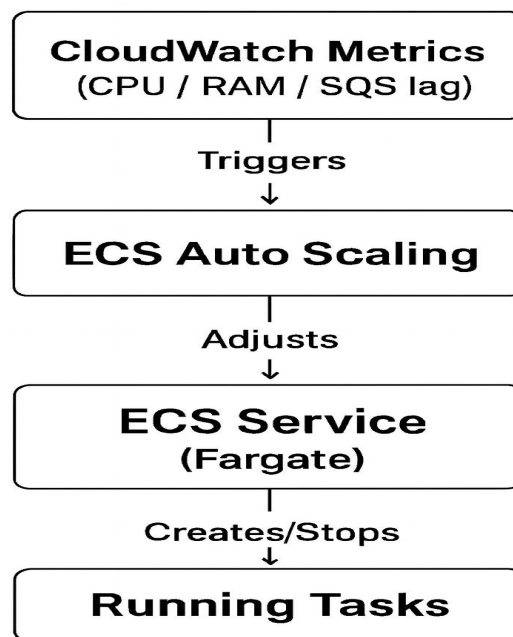
AWS manages:

- Kernel patching
- OS security
- SELinux isolation

Extra: Fargate tasks run on **dedicated kernel sandboxes**, reducing noisy-neighbor risks compared to EC2.

14. Auto Scaling Options

- ECS Service Auto Scaling (CPU/Memory/Requests)
- Event-driven scaling (SQS)
- KEDA for Kubernetes (EKS)



15. Fargate Spot (Cost Optimization)

Feature	On-Demand	Fargate Spot
Cost	Standard	Up to 70% cheaper
Interruptions	No	Yes (2-minute warning)
Best for	Production workloads	Fault-tolerant workloads (queues, retries)

AWS Fargate Spot is a low-cost compute option that lets you run interruption-tolerant ECS tasks on spare AWS capacity at up to 70% reduced cost.

16. Deployment Strategies

- Rolling Deployments
- Blue-Green (recommended)
- Canary Deployments

17. Monitoring & Troubleshooting

- CloudWatch Logs
- CloudWatch Metrics
- FireLens
- AWS X-Ray

Common Issue	Resolution
Task stuck in “Pending”	Increase ENI quota / check subnet
Image pull failure	Fix IAM or ECR access
Task exits immediately	Check logs / health checks
High latency	Check VPC routing / NAT Gateway bottlenecks

18. Reference Architecture

Users → ALB → ECS Service → Fargate Tasks → (Optional) EFS



CloudWatch Logs + AWS X-Ray

Optional: API Gateway → ALB → ECS Service → Fargate Tasks

19. Regions and Availability

- Fargate is available in most regions.
- Fargate for EKS is **not supported everywhere**.

20. When NOT to Use Fargate

Avoid Fargate if:

- You need GPU workloads
- You need daemon sets (EKS) like logging or monitoring agents
- You have very high traffic scale where EC2 Spot is cheaper long-term
- You require custom host-level configuration (kernel, drivers)
- You need predictable cost at massive scale (Reserved EC2 may be cheaper)

AWS Fargate Capacity Note

Category	Limit Type	Explanation
AWS Fargate	✗ No fixed task limit per Region	Capacity is based on vCPU quota , task size, available subnet IPs, and ENI limits.
AWS Fargate Spot	✗ No fixed limit	Availability depends on regional spare capacity ; workloads must tolerate interruptions.